



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2020 – 0262



Agricultural Machinery – Four-Wheel Tractor NEW HOLLAND TS6.120

Test Report Number	2020 – 0262
Date of Issuance	July 3, 2020
Valid Until	July 3, 2025
Agricultural Machinery	Four-Wheel Tractor
Type	Four-wheel drive; Four-cylinder, turbo-charged and intercooled diesel engine; 8F + 8R Synchromesh transmission system; Full hydrostatic steering system
Brand	NEW HOLLAND
Model	TS6.120
Test Requested by	FILLHOLLAND CORPORATION LCG Building, Osmeña St. Cagayan de Oro

SUMMARY

Performance Criteria	Requirement as per PAES 118:2001	AMTEC Test
Rated Maximum Power, kW	88.0 ^a	88.0 ^a
Rated Engine Shaft Speed, rpm	2200 ^a	2200 ^a
Maximum PTO Power, kW	70.9 ^a	74.9
Percentage PTO power of the Engine, %	80.6	85.1
At Maximum PTO Power		
Engine Speed, rpm	ND ^b	1825
PTO Shaft Speed, rpm	ND ^b	520
PTO Torque, N-m	ND ^b	1375.8
Fuel Consumption, L/h	ND ^b	29.8
Specific Fuel Consumption, g/kW-h	ND ^b	334.4
At 540 rpm PTO Shaft Speed		
Engine Speed, rpm	ND ^b	1891
PTO Torque, N-m	ND ^b	1308.0
PTO Power, kW	ND ^b	73.9
Fuel Consumption, L/h	ND ^b	30.6
Specific Fuel Consumption, g/kW-h	ND ^b	347.2

^a Data were obtained from operator's manual and nameplate.

^b Not specified by the manufacturer

Note: ND – No Data

SUMMARY (Cont'n)

Performance Criteria	Requirement as per PAES 118:2001	AMTEC Test
Main Gear	1,2,3,4 ^a	1,2,3,4
Range Gear	Low, High ^a	Low, High
Shuttle Gear	Forward, Reverse ^a	Forward, Reverse
Slowest Forward Speed, kph	2.8 ^a	3.18
Fastest Forward Speed, kph	29.6 ^a	32.4
Maximum Hydraulic Lift, kN	28.8 (2940.0 kg) ^a	32.3 ^d
Radius of Turning Circle, mm (Without Brakes)		
a) Right Turn	ND	5063
b) Left Turn	ND	5058
Position and/or Draft Implement Controls	Shall be provided	Provided
Three-Point Linkage	Category II	Category II
Power-Take-Off Shaft	540 rpm shall have 6 straight splines or 1000 rpm shall have 21 involute splines	540 and 1000
Tractor Speed with Implement	8 kph, maximum	NM
Wheel Tread Adjustment	Shall be provided	Provided
PTO Casing/Cover	Shall be provided	Provided
Roll-Over-Protective Structure (ROPS)	Shall be provided	Provided
Protective Seat Belt	Shall be provided	Provided

^a Data were obtained from operator's manual and nameplate.^dWith two (2) booster cylinders

Note: ND – No Data, NM – Not Measured

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous valid test reports of similar brand and model	2014-187
3. Manufacturer	CNH INDUSTRIAL, S.A. de C.V. Made in Mexico
4. Serial number	NH06999M
5. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	No Failure or abnormalities was observed.
6. Other remarks	No breakdown occurred

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Dates of Test	June 19, 23, & 24, 2020
Location of Test	AMTEC Test Laboratory, UPLB, College, Los Baños, Laguna
Items Tested	Operator's Manual, PTO Load Test, Hydraulic Lift, Travelling Speed, and Radius of Turning Circle/Area

METHODS OF TEST

The tractor was tested based on the PNS/PAES 119:2001 Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

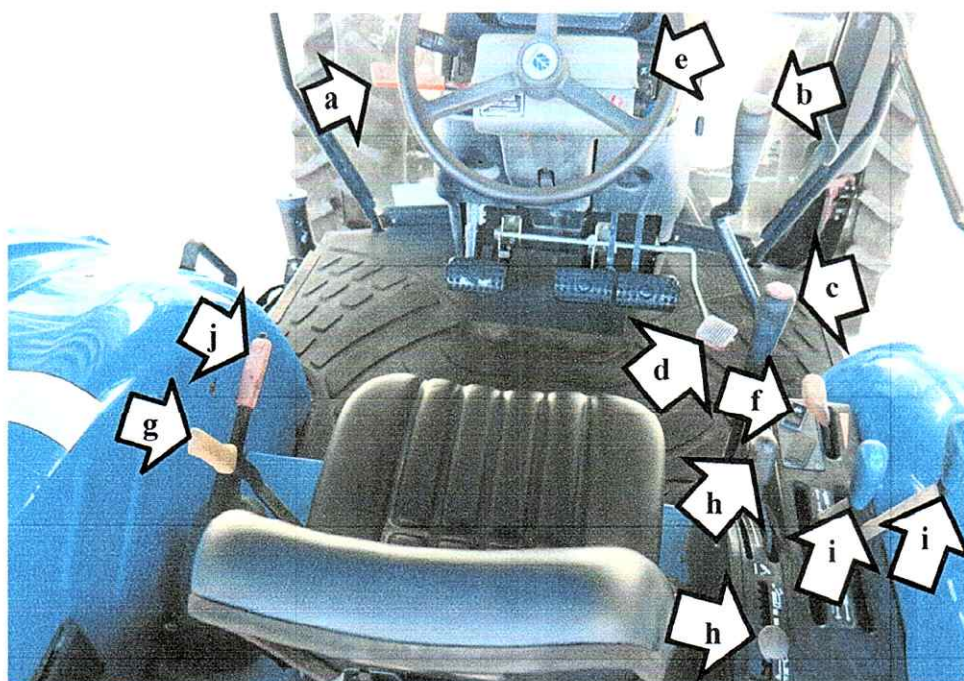


Figure 1. Controls of NEW HOLLAND TS6.120 Four-Wheel Tractor include the following:
(a) Shuttle gear shift lever, (b) Main gear shift lever, (c) Range gear shift lever, (d) Foot throttle pedal, (e) Front wheel drive control switch, (f) Differential lock control switch, (g) PTO clutch, (h) Three-point link hydraulic control lever, (i) Auxilliary hydraulic control levers, and (j) Parking brake lever.

SPECIFICATIONS

Item		Manufacturer's Specification^a	AMTEC Verification
1	Overall dimensions and weight of tractor		
1.1	Length, mm	4030	4410
1.2	Width, mm	2112	2160
1.3	Height, mm	2985 (ROPS)	3000 (ROPS)
1.4	Dry weight of tractor, kg		
1.4.1	Front	ND	NM
1.4.2	Rear	ND	NM
1.4.3	Total	ND	NM
2	Engine		
2.1	Brand	IVECO Tier III F4CE9484J	IVECO Tier III F4CE9484J
2.2	Make	IVECO, Italy	IVECO, Italy
2.3	Model	F4CE	F4CE
2.4	Serial number	001452712	001452712
2.5	Output power, kW	88.0	88.0
2.6	Rated shaft speed, rpm	2200	2200
2.7	Number of cylinders	4	4
2.8	Bore x stroke, mm	104 x 132	104 x 132
2.9	Displacement, cm ³	4500	4500
2.10	Type	Four-stroke cycle, Four-cylinder, Turbo-charged and Intercooled, Diesel	Four-stroke cycle, Four-cylinder, Turbo-charged and Intercooled, Diesel
2.11	Fuel used	Diesel	Diesel
2.12	Governor	ND	ND
2.13	Air cleaner	Dry type	Dry (double element)
2.14	Lubrication system	Force-feed	ND
2.15	Cooling system	Water-cooled	Water-cooled
2.16	Starting system	Electric starter	Electric starter
2.17	Electrical system	Battery-powered	Battery-powered
2.18	Exhaust system	Vertical muffler	Vertical muffler
3	Steering system	Full hydrostatic	Full hydrostatic

^aData were obtained from operator's manual and nameplate

Note: ND - No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification^a	AMTEC Verification
4	Power take-off		
4.1	Location	Rear of the tractor	Rear of the tractor
4.2	Type	Type I	Type I
4.3	Diameter of PTO shaft ends, mm	35	35
4.4	Number of splines	6 and 21 (interchangeable)	6 and 21 (interchangeable)
4.5	Rated shaft speed, rpm	540 and 1000	540 and 1000
4.6	Height above ground, mm	ND	750
4.8	Direction of rotation (viewed from the rear of the tractor)	Clockwise	Clockwise
4.9	Mode of operation	Lever-operated	Lever-operated
4.9.1	Speed change	540-1000 (through interchangeable PTO output shaft)	540-1000 (through interchangeable PTO output shaft)
4.9.2	Clutch	Independent	Independent
4.10	Rated PTO Power, kW	70.9	70.9
5	Ground clearance, mm	ND	380
6	Three-point linkage		
6.1	Category	II	II
6.2	Length of lift arms, mm	ND	260
6.3	Length of lower links, mm	ND	920
6.4	Distance of lift arm pivot point from rear-wheel axis, mm		
6.4.1	Horizontally	ND	NM
6.4.2	Vertically	ND	NM
6.5	Horizontal distance between the two lower links, mm	ND	610
6.6	Horizontal distance between the two lift arms endpoints, mm	ND	1100

^aData were obtained from operator's manual and nameplate

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification^a	AMTEC Verification
6.7	Length of upper link, mm		
6.7.1	Minimum	ND	680
6.7.2	Maximum	ND	860
6.8	Distance of upper link pivot point from rear wheel axis, mm		
6.8.1	Horizontally	ND	NM
6.8.2	Vertically	ND	NM
6.9	Distance of lower link pivot point from rear wheel axis, mm		
6.9.1	Horizontally	ND	NM
6.9.2	Vertically	ND	NM
6.10	Distance of lower link pivot point to lift rod pivot points on lower links, mm	ND	560
6.11	Length of lift rods, mm		
6.11.1	Minimum	ND	750
6.11.2	Maximum	ND	990
6.12	Height of lower hitch points relative to the rear-wheel axis, mm		
6.12.1	In lower position	ND	420
6.12.2	In high position	ND	565
6.13	Height above ground of lower hitch points when locked in transport position, mm	ND	1350
6.14	Diameter of top link pin, mm	ND	25.7
6.15	Diameter of lower link pin, mm	ND	28
7	Drawbar		
7.1	Type	ND	Swinging
7.2	Drawbar hole diameter, mm	ND	32
7.3	Drawbar thickness, mm		
7.3.1	Upper clevis	ND	25
7.3.2	Lower clevis	ND	39

^aData were obtained from operator's manual and nameplate.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
8	Transmission system		
8.1	Main clutch and PTO clutch	Independent	Independent
8.2	Transmission gears		
8.2.1	Type	Synchromesh	Synchromesh
8.2.2	Main gear	1,2,3,4	1,2,3,4
8.2.3	Range gear	Low, High	Low, High
8.2.4	Shuttle gear	Forward, Reverse	Forward, Reverse
8.3	Differential lock	Button	Button
9	Tire		
9.1	Type	Bias-ply	Bias-ply
9.2	Front tire size, mm	ND	379 x 610
9.3	Rear tire size, mm	ND	468 x 864
9.4	Wheel base, mm	2520	2510
9.5	Wheel tread		
9.5.1	Front tread, mm	ND	1670
9.5.2	Rear tread, mm	ND	1640
10	Brake system		
10.1	Type according to manner of applying brake	Multidisc, wet	ND
10.2	Type according to manner of transmitting the force from the controls	Hydraulics	Hydraulics
10.3	Parking brake	Mechanical	Mechanical

^aData were obtained from operator's manual and nameplate.

Note: ND – No Data, NM – Not Measured

RESULTS

Table 1. Radius of turning circle/area of the NEW HOLLAND TS6.120 Four-Wheel Tractor

Item	Without Brake		With Brake	
	Right Turn	Left Turn	Right Turn	Left Turn
Radius of Turning Circle, mm	5063	5058	4320	4475
Radius of Turning Area, mm	5215	5210	4473	4628

Table 2. Traveling speed of the NEW HOLLAND TS6.120 Four-Wheel Tractor at different gear settings

Range Gear Settings	Main Gear Settings	Shuttle Gear Settings	
		Forward (kph)	Reverse (kph)
Low	First	3.18	3.33
	Second	4.91	5.18
	Third	6.73	7.09
	Fourth	9.27	9.80
High	First	11.7	12.5
	Second	17.8	19.1
	Third	23.8	26.0
	Fourth	32.4	34.0

RESULTS (Cont'n)

Ambient Condition

Temperature, °C : 31.0
 Relative Humidity, % : 73.9

Table 3. Power Take-Off Load Test of the NEW HOLLAND TS6.120 Four-Wheel Tractor

SPEED		PTO		Fuel Consumption (L/h)	Specific Fuel Consumption (g/kW-h)
Engine (rpm)	PTO (rpm)	Torque (N-m)	Power (kW)		
2553	730	62.7	4.79	10.6	1849.6
2519	720	270.5	20.4	13.6	561.7
2446	700	594.0	43.5	22.1	427.0
2369	680	994.0	70.7	33.0	391.8
2307	660	1028.3	71.0	33.3	393.8
2223	640	1066.5	71.4	32.7	384.9
2166	620	1122.7	72.8	32.5	374.8
2091	600	1154.2	72.5	32.1	372.4
2027	580	1204.2	73.1	31.7	364.3
1952	560	1250.0	73.3	31.0	355.7
1891	540	1308.0	73.9	30.6	347.2
1825	520	1375.8	74.9	29.8	334.4
1751	500	1407.0	73.6	29.3	334.0
1679	480	1463.8	73.5	29.1	332.1
1613	460	1541.3	74.2	28.6	324.3
1547	440	1609.7	74.1	27.5	311.3
1403	400	1740.2	72.8	26.8	309.5

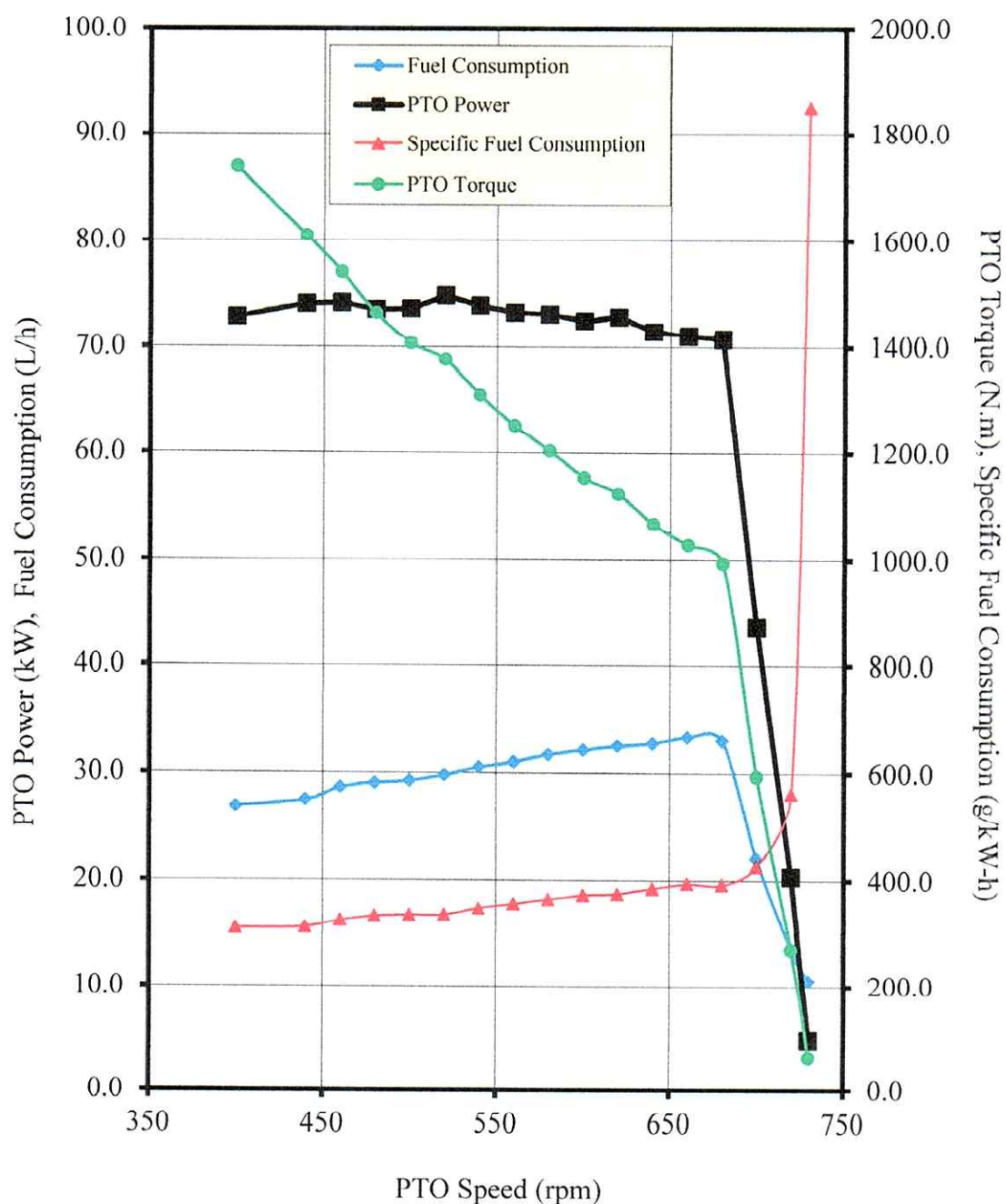
RESULTS (Cont'n)

Figure 3. Performance characteristic curves of the
NEW HOLLAND T6.120 Four-Wheel Tractor

RESULTS (Cont'n)

Figure 2. PTO test set-up of the NEW HOLLAND TS6.120 Four-Wheel Tractor

RESULTS (Cont'n)

Table 4. Hydraulic Lift Test Data of the NEW HOLLAND TS6.120 Four-Wheel Tractor

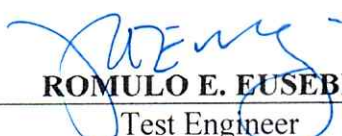
Trial	On the Frame (610mm from lower link end)	
	kg	kN
1	3290.7	32.3
2	3288.7	32.3
3	3286.1	32.2
4	3286.9	32.2
5	3296.7	32.3
Average	3289.8	32.3

Note: Two booster cylinder applied



Figure 4. The NEW HOLLAND TS6.120 Four-Wheel Tractor during the Hydraulic Lift test

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AMTEC Director**JUL 02 2020**

Date

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